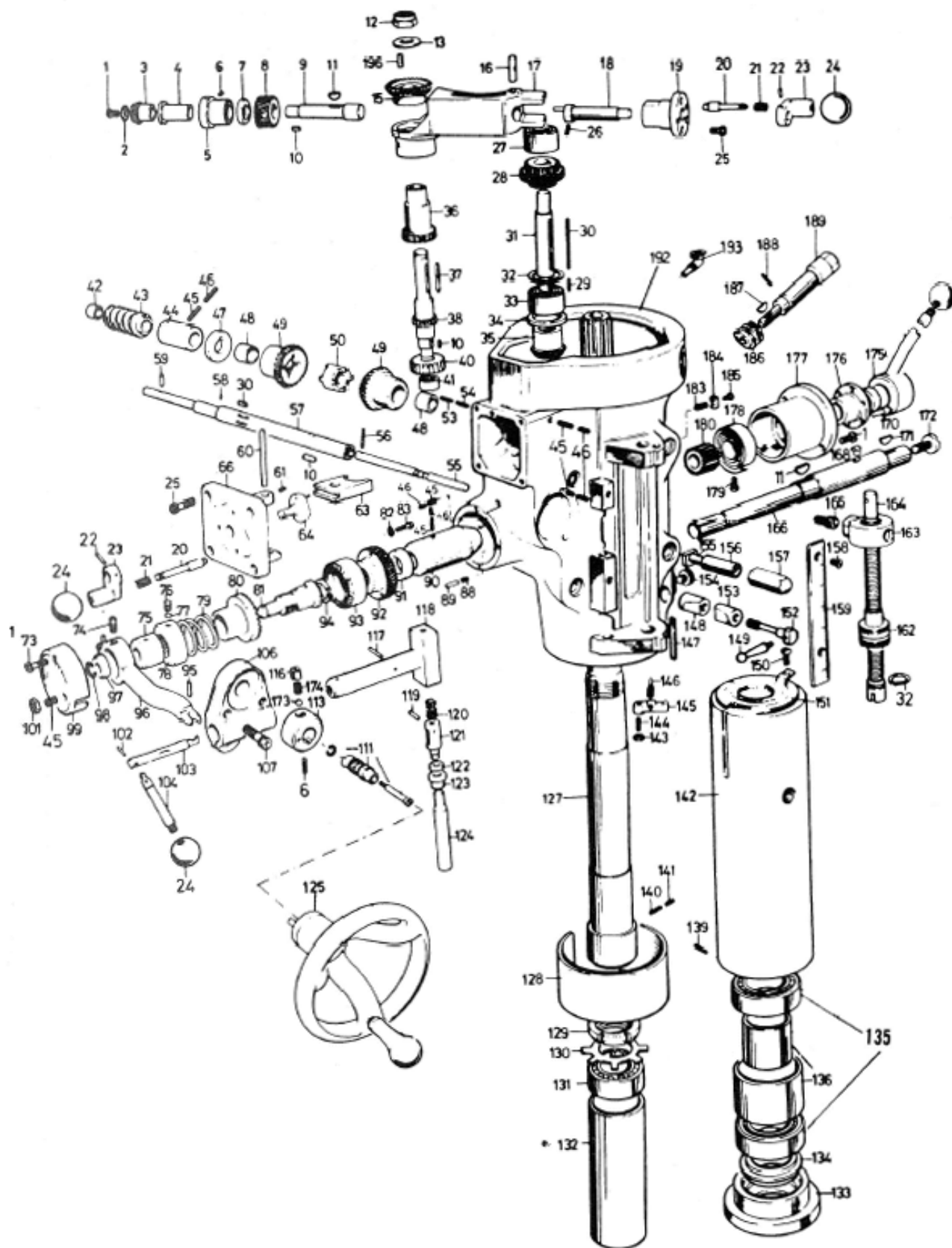


# 2J- AND J-HEAD LOWER HOUSING



TP5343

Figure 4.5 - 2J- and J-Head Lower Housing Assembly

## 2J- AND J-HEAD LOWER HOUSING

| ITEM | CODE NO. | DESCRIPTION                                | QTY |
|------|----------|--------------------------------------------|-----|
| 1    | 11011445 | Screw, Round Hd Machine, #10-24 x .375" Lg | 3   |
| 2    | 12190163 | Washer, Bevel Pinion                       | 1   |
| 3    | 12190203 | Gear, Feed Bevel Pinion                    | 1   |
| 4    | 12190164 | Sleeve, Feed Worm Gear Shaft               | 1   |
| 5    | 11192303 | Bushing, Worm Cradle                       | 1   |
| 6    | 11011287 | Screw, Stainless Steel, .250"-20 x .250"   | 2   |
| 7    | 12190165 | Spacer, Worm Gear                          | 1   |
| 8    | 12190166 | Gear, Feed Drive Worm                      | 1   |
| 9    | 12190167 | Shaft, Feed Drive Worm Gear                | 1   |
| *    | 12193440 | Assembly, Gear Drive Shaft                 | 1   |
| 10*  | 12190162 | Key, Worm Shaft, Sq, .125" x .312"         | 3   |
| 11   | 11013078 | Key, Woodruff #7                           | 3   |
| 12   | 11191796 | Locknut, Flexlok, .375"-24                 | 1   |
| 13   | 12190199 | Washer, .375"                              | 1   |
| 15   | 11192209 | Gear, Feed Reverse Bevel                   | 1   |
| 16   | 12190168 | Pin, Feed Engage                           | 1   |
| 17   | 12190059 | Cradle, Worm Gear                          | 1   |
| 18   | 12190169 | Throw-Out, Worm Gear Cradle                | 1   |
| 19   | 12190170 | Sleeve, Shift                              | 1   |
| 20   | 12190138 | Pin, Shift                                 | 2   |
| 21   | 11192052 | Spring, Compression                        | 2   |
| 22   | 11010517 | Pin, Roll, .125" x .750"                   | 2   |
| 23   | 12190064 | Crank, Shift (see 12193443)                | 1   |
| 24   | 11192151 | Knob, Bakelite, .250"-20                   | 4   |
| 25   | 11011010 | Screw, Socket Hd Cap, #10-24 x .500" Lg    | 7   |
| 26   | 11011258 | Screw, Set, #10-24 x .375"                 | 1   |
| 27   | 12190181 | Bushing, Cluster Gear Shaft Upper          | 1   |
| 28   | 28007099 | Assembly, Cluster Gear (see 12193504)      | 1   |
| 29   | 12190148 | Key, Sq, .125" x 1.750"                    | 1   |
| *    | 12193544 | Assembly, Bevel Feed Pinion                | 1   |
| 30*  | 12190175 | Assembly, Key, Sq, .125" x .563"           | 2   |
| 31   | 28007106 | Shaft, Cluster Gear                        | 1   |
| 32** | 11190836 | Ring, External Retaining, #5100-62         | 2   |
| 33** | 12190149 | Bearing, Bevel Gear                        | 1   |
| 34** | 12190150 | Spacer, Bevel Gear Thrust                  | 1   |
| 35** | 12190180 | Pinion, Feed Reverse Bevel                 | 1   |
| 36*  | 12190146 | Gear, Feed Driving                         | 1   |
| 37*  | 12190176 | Key, Round End, Sq, .125" x .750"          | 1   |
| 38*  | 12190145 | Shaft, Cluster Gear Input                  | 1   |
| 40*  | 12190144 | Gear, Feed Drive                           | 1   |
| 41   | 11190310 | Bearing, Needle, Torrington #B-66          | 1   |
| 42   | 11193637 | Bushing                                    | 1   |
| 43   | 28007307 | Gear, Worm Speed Control                   | 1   |
| 44   | 12190155 | Bushing, Feed Worm Shaft                   | 1   |
| 45   | 11011268 | Screw, Stainless Steel, Cup                | 6   |
| 46   | 11011542 | Screw, Set, .312"-18 x .938"               | 5   |
| 47   | 11190152 | Washer, Feed Worm Shaft Thrust             | 1   |

| ITEM | CODE NO. | DESCRIPTION                                    | QTY |
|------|----------|------------------------------------------------|-----|
| *    | 12193435 | Assembly, Bevel Feed Gear                      | 1   |
| 48*  | 11183646 | Bushing, Bronze, .502" x .628" x 5.00"         | 2   |
| 49*  | 12190151 | Gear, Feed Reverse Bevel                       | 2   |
| 50   | 12190153 | Clutch, Feed Reverse                           | 1   |
| 53   | 11011547 | Lock, Screw, Stainless Steel, .312"-18 x .156" | 1   |
| 54   | 11011375 | Screw, Stainless Steel, .312"-18 x .312"       | 1   |
| 55   | 12190157 | Rod, Reverse Clutch                            | 1   |
| 56   | 11010509 | Pin, Roll, .093" x .750"                       | 1   |
| 57   | 12190198 | Shaft, Feed Worm                               | 1   |
| 58   | 12190200 | Pin, .093" x .312" Lg                          | 1   |
| 59   | 28007308 | Pin, .100 x .438" Lg                           | 1   |
| 60   | 12190179 | Rod, Feed Shift                                | 1   |
| 61   | 11011260 | Screw, Stainless Steel, #10-32 UNF x .250"     | 1   |
| 63   | 11190061 | Fork, Feed Gear Shift                          | 1   |
| 64   | 12193446 | Assembly, Cluster Gear Shift Crank             | 1   |
| 66   | 12190065 | Cover, Cluster Gear                            | 1   |
| 73   | 11011014 | Screw, Cap, #10-24 x 1.500" Lg                 | 2   |
| 74   | 12190188 | Pin, Stop                                      | 2   |
| 75   | 12190098 | Ring, Clutch                                   | 1   |
| 76   | 11011265 | Screw, Stainless Steel, .250"-20 x .250" Lg    | 1   |
| 77   | 12190073 | Plug, Brass, .187"Ø x .093"                    | 1   |
| 78   | 12190105 | Locknut, Overload Clutch                       | 1   |
| 79   | 11192055 | Spring, Safety Clutch                          | 1   |
| 80   | 11192302 | Clutch, Overload                               | 1   |
| 81   | 12193549 | Sleeve, Overload Clutch                        | 1   |
| 82   | 11191920 | Washer, Single Spring                          | 3   |
| 83   | 11011431 | Screw, Round Hd Machine, #8-32 x .625"         | 3   |
| 88   | 11192032 | Spring, Compression, .250"Ø x 1.250"           | 1   |
| 89   | 12190096 | Plunger, Overload Clutch Lever Spring          | 1   |
| 90   | 12190106 | Bushing, Quill Pinion Shaft                    | 1   |
| 91   | 12190104 | Spacer, Pinion Shaft Worm Gear                 | 1   |
| 92   | 11190103 | Gear, Overload Clutch Worm                     | 1   |
| 93   | 12190102 | Ring, Overload Clutch                          | 1   |
| 94   | 11190870 | Ring, External Retaining                       | 1   |
| 95   | 11010717 | Pin, Dowel, .187" x .750"                      | 1   |
| 96   | 12193427 | Assembly, Overload Clutch Trip Lever           | 1   |
| 97   | 12190097 | Washer, Overload Clutch                        | 1   |
| 98   | 11190822 | Ring, External Retaining, #5100-37             | 1   |
| 99   | 12190068 | Cover, Clutch Arm                              | 1   |
| 101  | 11011740 | Nut, Hex Jam, .250"-20                         | 1   |
| 102  | 11010717 | Pin, Dowel, .187" x .750"                      | 1   |
| 103  | 12190094 | Rod, Cam                                       | 1   |
| 104  | 12190095 | Handle, Trip                                   | 1   |
| 106  | 12190067 | Bracket, Feed Trip                             | 1   |
| 107  | 11011035 | Screw, Socket Hd Cap, .250"-20 x 1.00"         | 1   |
| 111  | 12193433 | Assembly, Reverse Knob                         | 2   |
| 113  | 12190159 | Assembly, Handwheel Clutch (see 12193519)      | 1   |
| 116  | 12190154 | Assembly, Handwheel Clutch Spring              | 1   |
| 117  | 11010515 | Pin, Roll, .125" x .562"                       | 1   |
| 118  | 12190093 | Assembly, Cam Rod Sleeve                       | 1   |

| ITEM | CODE NO. | DESCRIPTION                                       | QTY |
|------|----------|---------------------------------------------------|-----|
| 119  | 11010513 | Pin, Roll, .125" x .437"                          | 1   |
| 120  | 11192053 | Spring, Compression                               | 1   |
| 121  | 12190091 | Plunger, Trip                                     | 1   |
| 122  | 12190092 | Bushing, Feed Trip Plunger                        | 1   |
| 123  | 12190090 | Bushing, Trip Plunger                             | 1   |
| 125  | 12190089 | Plunger, Feed Trip                                | 1   |
| 125  | 12193519 | Assembly, Handwheel                               | 1   |
| 127  | 12190191 | Spindle                                           | 1   |
| 128  | 11190081 | Skirt, Quill                                      | 1   |
| 129  | 11191790 | Locknut, #06                                      | 1   |
| 130  | 11191942 | Lockwasher, #W-06                                 | 1   |
| 131  | 11190237 | Bearing, Spindle                                  | 1   |
| 132  | 12190197 | Sleeve, Bearing                                   | 1   |
| 133  | 12190196 | Nosepiece                                         | 1   |
| 134  | 12780915 | Shield, Spindle Dirt                              | 1   |
| 135  | 11190238 | Bearing, Spindle, Set                             | 1   |
| 136  | 12193513 | Bearing, Spindle, Set                             | 1   |
| 139  | 11011265 | Screw, Stainless Steel, .250"-20 x .250" Lg       | 1   |
| 140  | 12193540 | Screw, Set, R-8 Collet                            | 1   |
| 141  | 11011545 | Screw, Locking Set, .250"-32 x .125"              | 1   |
| 142* | 12190192 | Quill (See 12194541)                              | 1   |
| 143  | 28300336 | Nut, Steel, #6-32 NC                              | 1   |
| 144  | 28300609 | Screw, Stainless Steel, #6-32 x .750"             | 1   |
| 145  | 28007042 | Lever, Feed Trip (see 12193498)                   | 1   |
| 146  | 12190185 | Pin, Trip Lever                                   | 1   |
| 147  | 12200103 | Rod, Indicator                                    | 1   |
| 148  | 12190109 | Sleeve, Quill Lock, Tapped                        | 1   |
| 149  | 12200098 | Handle, Lock                                      | 2   |
| 150  | 11011595 | Screw, Washer Hd Machine, #10-32 x .375"          | 1   |
| 151  | 11192403 | Strainer, Felt                                    | 1   |
| 152  | 12190111 | Bolt, Quill Lock                                  | 1   |
| 153  | 12190110 | Sleeve, Quill Lock, Untapped                      | 1   |
| 154  | 12200102 | Screw, Rod Indicator Thumb                        | 4   |
| 155  | 12191620 | Bolt, Tee, .500"                                  | 2   |
| 156  | 12190135 | Spacer, Lower Clamping Bolt                       | 2   |
| 157  | 12191736 | Nut, Hex, .500" x 1.500"                          | 2   |
| 158  | 11011411 | Screw, Chem Black Round Hd Machine, #6-32 x .250" | 1   |
| 159  | 11195306 | Scale, Quill Micrometer Inch                      | 1   |
| 162  | 12190344 | Assembly, Quick Nut                               | 1   |
| 163  | 12190082 | Nut, Quill Stop                                   | 1   |
| 164  | 12190083 | Screw, Quill Stop Micro                           | 1   |
| 165  | 11011090 | Screw, Socket Hd Cap, .375"-24 x .625" Lg         | 1   |
| 166  | 28007063 | Shaft, Quill Pinion                               | 1   |
| 168  | 12200111 | Pin, Spring                                       | 1   |
| 170  | 11010541 | Pin, Roll, .187" x .750" Lg                       | 1   |
| 171  | 11013076 | Key, Woodruff #3                                  | 1   |
| 172  | 12190182 | Screw, Pinion Shaft Hub                           | 1   |
| 173  | 11192165 | Ball, Steel                                       | 1   |
| 174  | 11192054 | Spring, Compression                               | 1   |
| 175  | 12201033 | Assembly, Quill Feed Handle                       | 1   |

| ITEM   | CODE NO. | DESCRIPTION                                  | QTY |
|--------|----------|----------------------------------------------|-----|
| 176    | 12190079 | Hub, Quill Pinion                            | 1   |
| 177*** | 12190066 | Cover, Spring                                | 1   |
| 178*** | 11192020 | Spring, Clock                                | 1   |
| 179*** | 28007150 | Pin, Outside Clock Spring                    | 1   |
| 180    | 28007064 | Pinion, Quill (see 12190108)                 | 1   |
| 183    | 12190085 | Lever, Reverse Trip Ball                     | 1   |
| 184    | 12190086 | Plunger, Feed Reverse Trip                   | 1   |
| 185    | 12190087 | Screw, Reverse Trip Ball Lever               | 1   |
| 186    | 11192207 | Gear, Worm                                   | 1   |
| 187    | 11013077 | Key, Woodruff #5                             | 1   |
| 188    | 11011370 | Screw, Socket Hd Set, .250"-UNC x 20 x .375" | 1   |
| 189    | 12190850 | Shaft, Adjustable Worm                       | 1   |
| 192*   | 12190051 | Housing, Quill (see 12194541)                | 1   |
| 193    | 11193111 | Cup, Oil                                     | 1   |
| 196    | 12190162 | Key, Worm Shaft, Sq, .125" x 5/16"           | 1   |

\*Items 10, 36, 37 and 40 sold as assembly 12193440.

\*\*Items 32, 33, 34 and 35 sold as assembly 12193544.

\*\*\*Items 177, 178 and 179 sold as assembly 12193437